

Rib stress injuries in the 2012–2016 (Rio) Olympiad: a cohort study of 151 Australian Rowing Team athletes for 88 773 athlete days

Rachel Harris , Larissa Trease ,^{1,2} Kellie Wilkie,³ Michael Drew^{4,5}

¹Orthopaedics ACT, Woden, Australian Capital Territory, Australia

²Centre for Health Care in Remote and Extreme Environments, School of Medicine, University of Tasmania, Hobart, Tasmania, Australia

³Bodysystem Physiotherapy, Hobart, Tasmania, Australia

⁴Department of Physiotherapy, Australian Institute of Sport, Canberra, Australian Capital Territory, Australia

⁵Australian Collaboration for Research into Injury in Sport and its Prevention (ACRISP), Federation University Australia, Ballarat, Victoria, Australia

Correspondence to

Dr Rachel Harris, TRIGG, WA 6954, Australia; admin@drharris.com

Accepted 31 December 2019

ABSTRACT

Aim To describe the demographics, frequency, location, imaging modality and clinician-identified factors of rib stress injury in a cohort of elite rowers over the Rio Olympiad (2012–2016).

Methods Analysis of prospectively recorded medical records for the Australian Rowing Team in 2013–2015 and the combined Australian Rowing Team and Olympic Shadow Squad in 2016, examining all rib stress injuries.

Results 19 rib stress injuries (12 reactions and 7 fractures) were identified among a cohort of 151 athletes and included 12 female and 7 male cases, 11 open weight, 8 lightweight, 12 scull and 7 sweep cases. The most common locations of injury identified by imaging, were the mid-axillary line and rib 6. Period prevalence varied from 4% to 15.4% and incidence ranged from 0.27 to 0.13 per 1000 athlete days. There were no significant differences in prevalence by sex, sweep versus scull or weight class. There was a statistically significant increase in incidence in the pre-Olympic year (2015, $p < 0.001$). MRI was the most commonly used modality for diagnosis. Stress fracture resulted in median 69 (IQR 56–157) and bone stress reaction resulted in 57 (IQR 45–78) days lost to full on water training.

Conclusions In our 4-year report of rib stress injury in elite rowing athletes, period prevalence was consistent with previous reports and time lost (median ~10 weeks) was greater than previously published literature. Rib stress injury limits training and performance in elite rowers and MRI should be considered as a first line investigation.

report a well-defined cohort of elite rowing athletes across a multiple year Olympic cycle.^{1 8}

This study provides the first detailed profile of rib stress injury sustained by elite rowers over a 4-year Olympic cycle. The demographic data of Australian Rowing Squad athletes sustaining rib stress injuries is described, including gender, sweep versus scull, weight class and timing in season. Period prevalence, year to year incidence, descriptive analysis of injury site and imaging modality used for diagnosis of rib stress injury is detailed. The time cost of rib stress injury and difference in time cost between reaction and fracture are discussed, and several clinical factors which may contribute to rib stress injury are discussed and suggested as future research topics.

METHODS

Population and study period

We reviewed the prospectively collected data relating to rib stress injury sustained by members of the senior Australian Rowing Squad during the Rio Olympiad (October 2012 to August 2016). In 2013–2015 all rib stress injuries in athletes selected to the Australian Rowing Team were included. As no athlete with rib stress injury was selected to the team in 2016, the cohort was expanded to include members of the Olympic Shadow Squad (incorporating athletes striving for late Olympic qualification, including lightweight), and all rib stress injury cases were included. Due to a lower level of injury surveillance in non-Olympic years (2013–2015), athletes who were not selected to the Australian Rowing Team and who sustained a rib stress injury, were not included in this study.

Data collection

Data were prospectively collected by rowing specific sports medicine practitioners (doctors and physiotherapists) in each Australian state or travelling team. Data were entered onto the Athlete Medical System, an electronic health record for Australian Olympic sport athletes, as kept by the Australian Institute of Sport. Individual athletes provide consent for the storage and use of their medical record, prior to their initial use of the Athlete Medical System.

Definitions

Rib stress injury was defined as chest wall pain in an athlete who was removed from on water rowing training for more than 24 hours and who had positive radiological findings of bone stress

INTRODUCTION

Rib stress injuries comprise approximately 9% of all rowing related injuries¹ and affect between 8% and 22% of elite level rowers.² Rowers have been reported to lose between 3 and 6 weeks of training with rib stress injury³; have complications including persistent pain and non-union, and the injury can be career-ending.⁴

Bone stress injuries are the failed adaptation of bone to repetitive loading, causing an imbalance of normal bone resorption and formation that occurs in bone remodelling.¹ This can result in a range of pathologies from stress reaction to more severe stress fracture.^{5 6} Rib stress injury is the term used to define bone stress injury of the rib.

The current published literature on rib stress injury is largely anecdotal and includes small case series; one prospective injury study in rowers did not report any rib stress injuries.⁷ No publications



© Author(s) (or their employer(s)) 2020. No commercial re-use. See rights and permissions. Published by BMJ.

To cite: Harris R, Trease L, Wilkie K, et al. *Br J Sports Med* Epub ahead of print: [please include Day Month Year]. doi:10.1136/bjsports-2019-101584

injury—including both rib stress fracture and rib stress reaction. Rib stress fracture was diagnosed when cortical breach was demonstrated on any form of imaging (ie, MRI, CT scan, ultrasound and bone scintigraphy). A rib stress reaction was diagnosed where there was bone oedema (T2 weighted MRI) or increased radioisotope uptake (scintigraphy), with no imaging evidence of cortical breach. Imaging modalities included MRI, CT scan, ultrasound and bone scintigraphy and the test selection depended on geographical location, medical practitioner's preference and clinical presentation. Each injury was examined and recorded by a Rowing Australia approved sports medicine practitioner (predominantly physicians or trainees from the Australasian College of Sport and Exercise Physicians, or titled sports physiotherapists).

Time loss for each injury was calculated, and monitoring was continuous, from injury onset to return to full on water training; including both missed days and modified training. Return to training was individualised by treating sports medicine practitioners in collaboration with the athlete and coach, and was based on pain, clinical improvement and upcoming competition.

Exposure was measured by calculation of athlete days, previously described as player days in rugby,⁹ with terminology modified for the rowing population. Athlete days were calculated by season length and the number of athletes followed for that season (eg, 20 athletes followed for a season of 50 days would equal 1000 athlete days). Non-training days were included in this calculation as they were considered scheduled rest days. Total athlete days were not reduced to account for missed training from other illness nor injury. Domestic and international rowing seasons were defined as 1 October to the national selection regatta in any year, and the period between the national selection regatta and the benchmark event in that year, respectively. Individual electronic written consent was obtained from each injured athlete, for their data to be included.

STATISTICAL ANALYSIS

Incidence and period prevalence were calculated for the two cohorts separately. Prevalence is defined as the proportion of the population that were affected by rib stress injury during the specified period¹⁰ and period prevalence (prevalence) was calculated by dividing the number of injured athletes during the surveillance period by the total number of athletes within the season.¹¹ Frequency was defined as the number of injury events recorded during the surveillance period. Injury incidence rates for each season, were calculated with the following formula: number of new injuries/exposure in athlete days $\times$ 1000. Statistically significant differences were determined when the 95% CI was $p < 0.05$. All statistical analyses were performed in Stata/IC (V.15.1).

RESULTS

Over the 4 years, 15 individual athletes suffered 19 rib stress injuries. Four athletes each suffered two distinct injuries, none of which were recurrence, at a different rib, site and year. In the Australian Rowing Squad, rib stress injury occurred 12 times in females and 7 in males. There were 11 open weight, 8 light-weight, 12 scull and 7 sweep cases, 12 reactions and 7 fractures in the cohort (table 1).

Demographics

Over the 4-year Olympiad, a total cohort of 151 individual athletes were selected to the Australian Rowing Team or were monitored in the Shadow Squad. There were 85 individual athletes in the 2013–2015 Australian Rowing Team. No new

Table 1 Demographics and rib stress injury in Australian Rowing Squad 2013–2016

Year	Individual athletes*	Monitored athletes	Male athletes	Male rib stress injury†	Female athletes	Female rib stress injury†	Open weight athletes	Open weight rib stress injury†	Light weight Athletes	Light weight rib stress injury†	Sweep athletes	Sweep rib stress injury†	Scull athletes	Scull rib stress injury†	Total rib stress injury†	Surveillance period athlete days
2013	50	50	30	2	20	0	31	1	19	1	30	1	20	1	2	335
2014	21	51	28	1	23	3	38	3	13	1	30	3	21	1	4	335
2015	14	52	28	3	24	5	38	4	14	4	27	6	25	2	8	341
Total 2013–2015	85	153	86	6	67	8	107	8	46	6	87	10	66	4	14	1011
2016	66†	117	70	1	47	4	83	3	34	2	80	2	37	3	5	318
Olympiad	151	270	156	7	114	12	190	11	80	8	167	12	103	7	19	1329

*Individual athlete is the first time an athlete appears in the Australian Rowing Squad.

†Number of rib stress injury in each time period.

#No new athletes were selected to the Australian Rowing Team in 2016.

Table 2 Incidence (n/1000 athlete days) and period prevalence (%) of rib stress injury

Year	N	Athlete days	Rib stress injury	Incidence (n/1000 athlete days)	Season period prevalence	Male	Female	Open weight	Light weight	Sweep	Scull
2013	50	16 750	2	0.12	4.00	*	*	*	*	*	*
2014	51	17 085	4	0.23	7.80	*	*	*	*	*	*
2015	52	17 732	8	0.45	15.40	*	*	*	*	*	*
2013–2015	153	51 567	14	0.27	9.15	6.98	11.94	7.48	13.04	11.49	6.06
2016	117	37 206	5	0.13	4.27	1.43	8.51	3.61	5.88	2.50	8.11
Total	270	88 773	19	0.2							

*Period prevalence was not calculated yearly for subgroups in 2013–2015.

athletes were selected to the Australian Rowing Team in 2016, however, 66 new athletes were monitored in the Shadow Squad. The demographics of the cohort varied considerably each year, due to multiple selection factors beyond control of the research group (table 1). The overall athlete days for the Australian Rowing Squad over the Olympiad was 88 773 days, covering 270 monitored athlete seasons.

Incidence and prevalence

Overall injury incidence for the Australian Rowing Team (2013–2015) was 0.2714 per 1000 athlete days, and for the Shadow Squad (2016) was 0.1344 per 1000 athlete days (table 2). Prevalence for the Australian Rowing Team was 9.15%, however, varied widely from 4% in 2013 to 15.4% in 2015. The prevalence of rib stress injury in the Shadow Squad in 2016 was 4.27%. Prevalence was the highest among lightweight athletes at 13.04% in the first 3 years of the Olympiad (table 2). The overall 4-year period prevalence of having sustained at least one rib stress injury was 9.9% (including all 151 athletes monitored from 2013 to 2016) with 2.7% having more than one (n=4) and 27% of athletes with one rib stress injury having a second in the Olympiad. There was no statistically significant difference in prevalence between subgroups, however in 2016, male to female met the a priori alpha level ($p=0.05$). Prevalence was highest in females (11.94%), lightweight (13.04%) and sweep (11.49%) athletes in the 2013–2015 Australian Rowing Team

and in females (8.51%), lightweight (5.88%) and scull (8.11%) athletes in the 2016 Shadow Squad. It was not possible to calculate athlete days reduction due to other illness and injury, and as such, real world incidence is likely to be higher due to a lower exposure.

Timing of injury

Through the Olympiad, 11 (58%) of injuries occurred during the domestic season and eight (42%) in the international season. The highest number of injuries occurred in the pre-Olympic year of 2015 with eight (42%) separate rib stress injuries ($p<0.001$).

Location of injury

The sixth rib was most frequently injured with nine separate cases, and ribs 5–7 made up 15 of the 19 injuries in this cohort (figure 1). The most common sites of rib stress injuries were posterior, axillary and anterolateral ribs. The most common injury location in sweep athletes was the mid-axillary region, and in sculling athletes, the anterolateral and posterior (figure 2).

Imaging modality

The imaging modalities used to diagnose rib stress injury were MRI, CT scan, ultrasound and bone scintigraphy. For initial diagnosis the imaging modality used were MRI only (n=7), combination of bone scan and CT (n=1), combination of MRI

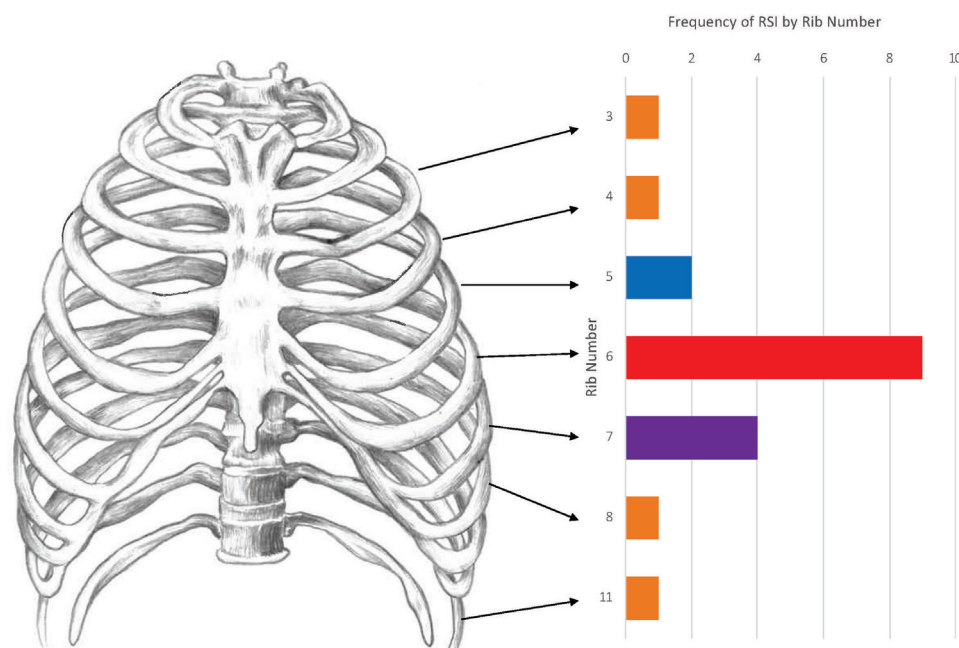


Figure 1 Rib stress injury (RSI) rib location and frequency.

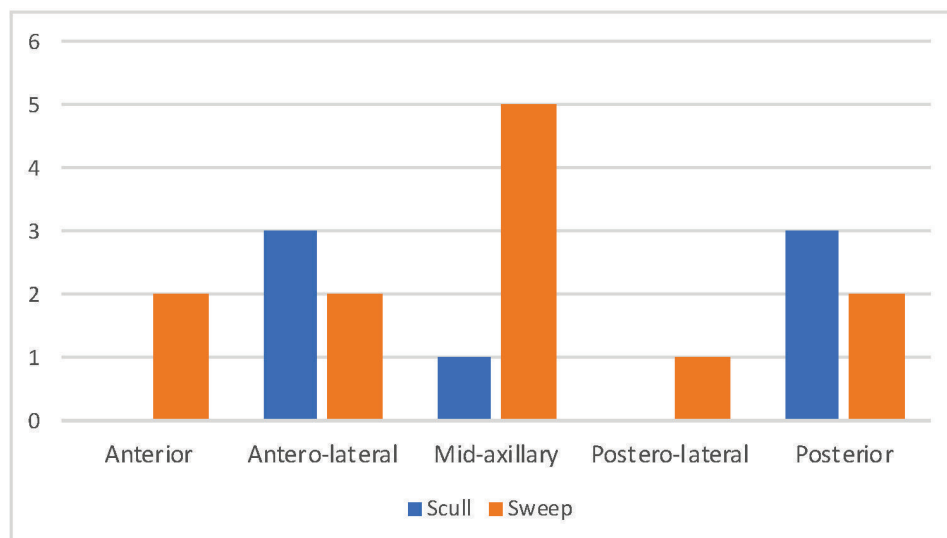


Figure 2 Location of rib stress injury in scull and sweep athletes.

and CT (n=5); MRI and bone scan (n=3); bone scan and ultrasound in combination (n=3). Ultrasound was used for diagnosis once, however, was frequently used for surveillance of injuries, when a Sport and Exercise Physician with ultrasound expertise, was working within the team environment.

Time cost associated with rib stress injury

The median time to return to full on water training was 69 days for fracture and 57 days for reaction. No athlete returned to training after sustaining a fracture before 51 days and all returned before 167 days. For athletes sustaining rib stress reaction, the quickest return to training was 24 days and all athletes returned to training before 97 days (figure 3).

Clinician identified factors contributing to rib stress injury in this cohort

Detailed analysis of factors which may have contributed to rib stress injury in each athlete was beyond the scope of this

research. For some cases, medical records yielded clinician identified factors that contributed to individual rib stress injury. In three athletes (16%), weight loss prior (intentional (n=2); illness-related (n=1)) to onset of symptoms was considered a likely contributing factor. In four female athletes (33% of female cases), amenorrhoea and low energy availability (LEA) were identified (two lightweight and two open weight) and one lightweight male athlete was identified to have LEA. Relative Energy Deficiency in Sport (RED-S), bone density, hormonal health and nutritional behaviours were an issue in eight athletes (42%). Five athletes (26%) had documented technique change or side change in the weeks leading up to their injury presentation, while six (32%) of the rib stress injuries followed marked load change, such as a training camp or return to training after week(s) off. Given the retrospective analysis, aetiological factors were not investigated and recorded uniformly across the cohort, and caution is advised interpreting these results.

DISCUSSION

This paper consolidates previous research on the demographics of rib stress injuries, establishes incidence and period prevalence across an Olympiad and highlights the significant training time cost of rib stress injury in an elite rowing programme. It highlights important clinical information associated with rib stress injuries including imaging modality and identifies areas for further research.

Incidence and period prevalence

Incidence ranged from 0.12 to 0.45 cases per 1000 athlete days. This incidence equates to 2–8 new cases of rib stress injury, in a squad of 50 rowers across a season of 340 training days. This is the first publication to establish incidence of rib stress injury reported in an international level rowing cohort over an Olympic cycle.

Prevalence of rib stress injury in the Australian Rowing Team and Shadow Squad was 8%–15% annually, consistent with previous reports.^{1 2 12 13} Prevalence was highest in females, lightweight and sweep athletes in the 2013–2015 Australian Rowing Team and in females, lightweight and scull athletes in the 2016 Shadow Squad, none of which reached statistical significance. The increased prevalence in female athletes in this cohort supports previous case reports^{2 14} and highlights the need for further research into gender specific factors using appropriate cohort designs. Conflicting studies, have either found no difference between sweep and

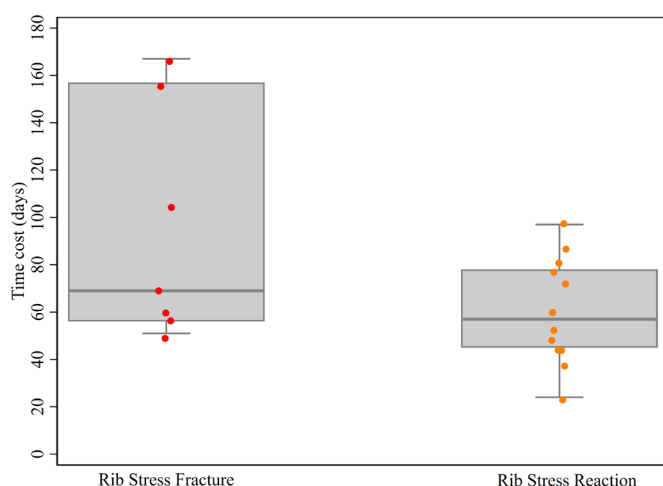


Figure 3 Time cost of rib stress injury in the Australian Rowing Squad over the Rio Olympiad. It is a box and whisker plot representing the distribution of the length of time between onset and return to full training of rib stress fracture and rib stress reactions. This represents median, 25th and 75th centiles, extremes of range and individual dots represent individual injuries.

scull injury prevalence¹⁵ or increased susceptibility in sweep athletes.^{12 16 17} Our research highlights the need for rib stress injury identification and prevention programmes in all rowers.

Time cost

This cohort confirms significant interruption to training and competition with rib stress injury.^{3 12 14 16 17} Across the Olympiad, no athlete with a rib stress injury medaled at a benchmark event, nor qualified for the Australian Rowing Team in the Olympic year. This contrasts previous reports which described athletes achieving sporting success soon after return to competition.^{16 18}

Median training time missed for rib stress fracture was 69 days and 57 for reaction. The greatest time loss due to fracture (167 and 157 days), accounts for two athletes that rowed through their injury due to the proximity of diagnoses to benchmark competition. This significantly prolonged recovery time and may highlight progression of injury due to sustained loading and whether cessation of rowing is required for rehabilitation in a timely manner. One athlete retired from rowing following their injury, and identified the injury as a contributing factor to this decision. The time cost of rib stress injury in our cohort are in excess of previously reported data^{4 18 19} which may be a consequence of authors varying definitions of a return to training.^{4 18 19} Regardless, it must be noted that no athlete sustaining reaction returned to full training before 24 days, and fracture 51 days, and highlights the importance of differentiation between fracture and reaction in the prediction of time cost.

Location of injury

The rowing injury literature repeatedly describes the most commonly injured ribs as 4–7.^{3 12 13 18} Our Australian cohort confirmed this finding and the most common sites for rib stress injury were antero-lateral (26%), mid-axillary (31%) and posterior (26 %) rib cage, with cortical breach occurring on the outer aspect of each rib fracture. Understanding the frequency of location may assist in further identification of the aetiology of rib stress injury, which continues to be the subject of debate.² The high number of posterior injuries in our cohort highlights the importance of clinical suspicion for posterior rib stress injury, which maybe under detected with the traditional chest wall testing paradigm^{3 20} and we caution clinicians working with rowers on the similarity in presentation between thoracic spine generated pain and posterior rib stress injury.

Imaging modality

There is no consensus in the literature regarding the most effective radiological investigation for the detection of rib stress injury.^{1 13 21 22} MRI alone was successfully used for detection in seven cases in this cohort. Where an injury was not clearly identified on MRI and there was high clinical suspicion, bone scintigraphy was performed. If cortical breach was not identified on MRI, targeted CT was used, due to the importance of distinction between reaction and fracture for prognosis of likely training time cost. Ultrasound was used successfully for rib stress injury monitoring, aligning with recommendations from Roston *et al*²² who proposed ultrasound as a useful modality for diagnosis and surveillance, when a skilled operator is available, and access to other imaging is limited. Based on our imaging experience across the Rio Olympiad, we recommend the use of MRI as a first line investigation, limiting potential radiation exposure to athletes without negative outcomes or delaying diagnosis.²³ The authors acknowledge limitations of MRI for identification of rib stress injury due to movement artefact, and the possibility of false negative scans. Rowing clinicians are reminded of the

importance of continued investigation with alternative imaging if there is a high clinical suspicion.

Clinician identified factors

Underlying intrinsic, training and equipment factors in athletes sustaining rib stress injury were not specifically recorded in the Athlete Medical System. However, analysis of the injury records identified specific risk factors that may have contributed to individual rib stress injury.

A number of health factors are known to contribute to bone stress injuries in athletes and these include RED-S, bone density, hormonal health and nutritional behaviours.^{1 3 8 24 25} Eight athletes (42%) in the Australian Rowing Squad were noted to have one or more of these risk factors, highlighting the need for further research into the association between RED-S, LEA and rib stress injury in rowing athletes.

Rib stress injury has been reported to increase following periods of heavy training or a dramatic increase in training load.^{12 17 26 27} Six (32%) rib stress injuries occurred following a marked increase in load, such as a training camp or return to training after week(s) off. The management of training load in athletes forms a critical component to injury prevention.²⁸ Change in technique or change in side has previously been suggested as a potential cause of rib stress injury.¹⁶ In the weeks leading up to diagnosis, five athletes (26%) in the Australian Rowing Squad were identified as having a focus on technique change, recent change in side or a switch from sweep to scull or vice versa. While clinician identified factors were not the aim of this research, they require further investigation to quantify their contribution to rib stress injury and help drive the development of preventative strategies. Further research must be directed toward the contribution of load and individual athlete health factors in the development of rib stress injury, if injury burden is to be reduced.

Limitations

Demographics

The demographics of the Australian Rowing Squad represent the real-world variation in yearly athlete performance and team selection. The Australian Rowing Team selection criteria changed annually through the Olympiad, and this, along with factors such as individual athlete performance and qualification of boats, mean that demographics across the Olympiad are not homogeneous. The authors recognise several limitations in this regard. The cohort was distinctly different in 2013–2015 and 2016. It would be ideal to extend the cohort in 2013–2015 to include all athletes who nominated for selection to the team. Unfortunately, this was not possible due to the large number of athletes, and the lack of standardised monitoring and sports medicine support across the country in non-Olympic years. We identify potential survivor bias in this cohort, as injured athletes may not nominate for selection for the following season, therefore reducing the number of overall rib stress injuries. This is an elite group of athletes, which reduces external validity in the subelite, under-age and recreational rowing community. Data were analysed retrospectively, meaning lost training days due to other illness and injury were not accounted for. This results in athlete days representing higher than actual exposure, and true incidence of rib stress injury likely underestimated.

CONCLUSION

Nineteen rib stress injuries were recorded in 15 elite rowers across an Olympiad, with a prevalence of 8%–15% and an incidence of 0.12–0.45 injuries per 1000 athlete days. No rib stress

Original research

fracture case returned to full training before 51 days and no reaction case before 24 days. The sixth rib is the most frequent site of injury and MR imaging is an important first line imaging modality for diagnosis of rib stress injuries. The pre-Olympic year showed a statistically significant increase in rib stress injuries numbers. This study may assist rowing clinicians form a basis of further research for preventative strategies, and best practice management of rib stress injuries in elite rowers.

What are the new findings?

- Time cost of rib stress injury is in excess of previous findings in the elite rower with fractures taking median of 69 days (IQR 56–157) to return to full training and reactions 57 days (IQR 45–78).
- Females have a tendency toward increased prevalence, but not statistically significant increase in risk of rib stress injury.
- No athlete with rib stress injury in the Olympiad won a medal in the year of their injury or at the Rio Olympics.

How might it impact on clinical practice in the future?

- MRI should be considered first line imaging for rib stress injury in elite athletes, to reduce lifetime radiation dose.
- The incidence and prevalence of rib stress injury is now established in Australian Olympic level rowers and this offers a baseline to measure the effectiveness of injury prevention interventions in all elite rowing programmes.

Twitter Rachel Harris @DrRachelHarris, Larissa Trease @DrLarissaTrease, Kellie Wilkie @KellieWilkie and Michael Drew @_mickdrew

Acknowledgements The authors would like to thank Rowing Australia and the Australian Institute of Sport for their support throughout the study. We acknowledge and thank the doctors and physiotherapists of the Australian Rowing Team and Australian provider network for their diligence in the care of Australian rowers and the completion of their medical records. The authors would like to thank Kyra Ginn for her illustration.

Contributors RH, LT and KW were involved in the original design, data collection, interpretation of the statistics and drafting and approving the manuscript. LT and KW were responsible for collected data from 2012–2016. MD undertook all statistical analyses and was involved in the interpretation of the statistics and drafting and approving the manuscript.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient and public involvement statement Rowers in the Australian Rowing Team were not involved in the development of the research questions or study design and the collection of this data did not demand any additional time or intervention burden for them. All members of the rowing community, including Australian Rowing Team athletes will benefit from the increased understanding of rib stress injuries in the sport and results of this study will be disseminated through the Australian Rowing Team medical staff to current athletes.

Patient consent for publication Not required.

Ethics approval Ethics approval was obtained from the Australian Institute of Sport Human Ethics Committee, Approval Number 20180403.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement No data are available. No data are available.

ORCID iDs

Rachel Harris <http://orcid.org/0000-0002-1955-894X>

Larissa Trease <http://orcid.org/0000-0002-8413-2814>

REFERENCES

- McDonnell LK, Hume PA, Nolte V. Rib stress fractures among rowers: definition, epidemiology, mechanisms, risk factors and effectiveness of injury prevention strategies. *Sports Med* 2011;41:853–901.
- Warden SJ, Gutschlag FR, Wajswelner H, et al. Aetiology of rib stress fractures in rowers. *Sports Med* 2002;32:819–36.
- Evans G, Redgrave A. Great Britain Rowing team guideline for diagnosis and management of rib stress injury: Part 1. *Br J Sports Med* 2016;50:266–9.
- Vinther A, Thornton JS. Management of rib pain in rowers: emerging issues. *Br J Sports Med* 2016;50:141–2.
- Hughes JM, Popp KL, Yanovich R, et al. The role of adaptive bone formation in the etiology of stress fracture. *Exp Biol Med* 2017;242:897–906.
- Pepper M, Akuthota V, McCarty EC. The pathophysiology of stress fractures. *Clin Sports Med* 2006;25:1–16.
- Wilson F, Gissane C, Gormley J, et al. A 12-month prospective cohort study of injury in international rowers. *Br J Sports Med* 2010;44:207–14.
- Thornton JS, Vinther A. Prevention of rib stress injury in rowers. what do we know and where do we need to go? *Sports Orthopaedics and Traumatology* 2018;34:278–86.
- Schwellnus M, Derman W, Page T, et al. Illness during the 2010 Super 14 Rugby Union tournament - a prospective study involving 22 676 player days. *Br J Sports Med* 2012;46:499–504.
- Gordis L. *Epidemiology*. 5th ed. Saint Louis: Elsevier Health Sciences, 2014.
- Knowles SB, Marshall SW, Guskiewicz KM. Issues in estimating risks and rates in sports injury research. *J Athl Train* 2006;41:207–15.
- Christiansen E, Kanstrup IL. Increased risk of stress fractures of the ribs in elite rowers. *Scand J Med Sci Sports* 1997;7:49–52.
- Dragoni S, Giombini A, Di Cesare A, et al. Stress fractures of the ribs in elite competitive rowers: a report of nine cases. *Skeletal Radiol* 2007;36:951–4.
- Hickey GJ, Fricker PA, McDonald WA. Injuries to elite rowers over a 10-yr period. *Med Sci Sports Exerc* 1997;29:1567–72.
- Rowing HJA. Rowing. In: Drinkwater BL, ed. *Women in sport*. Wiley, 2000: 486–93.
- Bojanić I, Desnica N. Stress fracture of the sixth rib in an elite athlete. *Croat Med J* 1998;39:458–60.
- Holden DL, Jackson DW. Stress fracture of the ribs in female rowers. *Am J Sports Med* 1985;13:342–8.
- Karlson KA. Rib stress fractures in elite rowers. A case series and proposed mechanism. *Am J Sports Med* 1998;26:516–9.
- Evans G, Redgrave A. Great Britain Rowing Team Guideline for Diagnosis and Management of Rib Stress Injury: Part 2 - The Guideline itself. *Br J Sports Med* 2016;50:270–2.
- Hooper I, Blanch P, Sternfeldt J. The development of a clinical management pathway for chest wall pain in elite rowers. *Journal of Science and Medicine in Sport* 2011;14:e104.
- Smoljanovic T, Bojanic I. Ewing sarcoma of the rib in a rower: a case report. *Clin J Sport Med* 2007;17:510–2.
- Roston AT, Wilkinson M, Forster BB. Imaging of rib stress fractures in elite rowers: the promise of ultrasound? *Br J Sports Med* 2017;51:1093–7.
- Cross TM, Smart RC, Thomson JEM. Exposure to diagnostic ionizing radiation in sports medicine: assessing and monitoring the risk. *Clinical Journal of Sport Medicine* 2003;13:164–70.
- Mountjoy M, Sundgot-Borgen J, Burke L, et al. The IOC consensus statement: beyond the Female Athlete Triad—Relative Energy Deficiency in Sport (RED-S). *Br J Sports Med* 2014;48:491–7.
- Hayashi D, Jarraya M, Engebretsen L, et al. Epidemiology of imaging-detected bone stress injuries in athletes participating in the Rio de Janeiro 2016 summer Olympics. *Br J Sports Med* 2018;52:470–4.
- Hosea TM, Hannafin JA. Rowing injuries. *Sports Health* 2012;4:236–45.
- Verrall G, Darcey A. Lower back injuries in rowing national level compared to international level rowers. *Asian J Sports Med* 2014;5:e24293.
- Raysmith BP, Drew MK. Performance success or failure is influenced by weeks lost to injury and illness in elite Australian track and field athletes: a 5-year prospective study. *J Sci Med Sport* 2016;19:778–83.